ETCNet: Encrypted Traffic Classification using Siamese Convolutional Networks

Lu Xu
New York University
New York, USA
lx643@nyu.edu

Daihui Dou
New York University
New York, USA
dd2956@nyu.edu

H. Jonathan Chao
New York University
New York, USA
chao@nyu.edu

ABSTRACT

As more and more Internet traffic is encrypted, classifying their flows for the usage in application-aware networking (AAN) and application-network integration (ANI) becomes increasingly important and challenging. Traditional deep packet inspection approaches are no longer capable of identifying the encrypted packet streams, and hence new traffic classification methods based on machine learning have recently been explored by several researchers. One major challenge of using machine learning to classify encrypted traffic is lacking real datasets. Collecting Internet traffic may leak users' sensitive information, which prohibits the network community from sharing the datasets they collected. In this poster, we propose ETCNet which is a model based on Siamese convolutional network to solve this issue. Our evaluation for the ETCNet shows that it can achieve high accuracy by only using 40 flows of each application to train it.

CCS CONCEPTS

• **Networks** → **Packet classification**; *Network monitoring*; • **Security and privacy** → Cryptanalysis and other attacks.

KEYWORDS

encrypted traffic classification, siamese convolutional network, machine learning

ACM Reference Format:
Lu Xu, Daihui Dou, and H. Jonathan Chao. 2020. ETCNet: Encrypted Traffic Classification using Siamese Convolutional Networks. In *Workshop on Network Application Integration/CoDesign (NAI'20), August 14, 2020, Virtual Event, USA.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3405672.3409492

1 INTRODUCTION

Internet traffic classification has become an important research area in recent years [5]. The ability to categorize traffic to its application is critical for application-aware networking (AAN) and application-network integration (ANI). By classifying each flow's application in real-time, routers or middleboxes are able to apply different scheduling or buffer management strategies to the flows that are sensitive to delay or packet loss, achieving better network performance and users' quality of experience (QoE). Moreover, traffic classification can also distinguish malware traffic from normal traffic [6] to enhance network security.

An encrypted Internet protocol, HTTP2, has been widely used by most top websites in the world [3]. According to [2], the usage of HTTP2 for the websites has reached 43.6%. Just last year alone, the adoption of HTTP2 increased by nearly 10%. The popularity of HTTP2 shows the necessity of developing an effective traffic classification scheme for encrypted packet streams. However, the shift has created tremendous difficulty for traffic classification, because the information in packet payload is no longer available for inspection.

While researchers are using machine learning to perform traffic classification, it has been very challenging for them to devise effective algorithms due to the lack of available datasets to train and evaluate their algorithms. Unlike many other machine learning problems, such as computer vision (CV) and natural language processing (NLP), network traffic contains users' personal information, making it not always collectable. Moreover, because of the sensitivity of personal data, researchers in the network community have the concern of sharing the real users' data even after they manage to collect them, making it very difficult, if not impossible, for the network community to collaboratively form a large dataset. What's more, the contents of Internet traffic tends to change quickly, rendering the collected dataset obsolete quickly and requiring a new effort of collecting them again.

To meet the need of classifying traffic and overcoming the problem of lacking datasets, we introduce ETCNet, a classification model based on Siamese convolutional networks (to be discussed in Section II), which only needs a small dataset to train the model while achieving high classification accuracy on encrypted Internet traffic. Heuristically, the model converts a multi-class classification problem to a binary classification problem, which reduces the problem difficulty and the datasets needed. The Siamese convolutional network takes a pair of traffic flows as input and produces an intermediate feature, capturing the level of similarity of the two flows. A classifier is then used to produce a binary label, indicating whether the two flows belong to the same application. To ensure the robustness of the model, not being influenced by spoofed identification or port numbers, the model only uses the encrypted payload of the packet stream. A detailed description of the feature selection is discussed in Section III.

To evaluate the ETCNet's prediction accuracy, we created a dataset via mocking the users' browsing behavior. As shown in Tab. 1, eight of the world's top one hundred applications [1] are chosen and all the applications use the HTTP2 protocol. We recorded the

NAI'20, August 14, 2020, Virtual Event, USA

ACM ISBN 978-1-4503-8044-7/20/08...$15.00
https://doi.org/10.1145/3405672.3409492

Table 1: The applications collected in our dataset

Reddit	Twitter	Spotify	Facebook
Instagram	Wikipedia	Amazon	Stackoverflow

packets passing through the Network Interface Card (NIC) while the mocked user is browsing those applications. We tested our model on the dataset which achieves an accuracy of 92%.

The main contributions of our work are summarized as follows:

- Our proposed ETCNet model for encrypted traffic classification is able to classify packet stream's applications by using a very small training dataset (e.g., only 40 flows for each application).
- We collected our own dataset for HTTP2 flows, more than 1500 flows for each application. We filtered raw Internet traffic and stored each flow to an individual file so that the dataset can be easily used by other researchers without much pre-processing needed. We have uploaded our dataset on Google Drive [1].

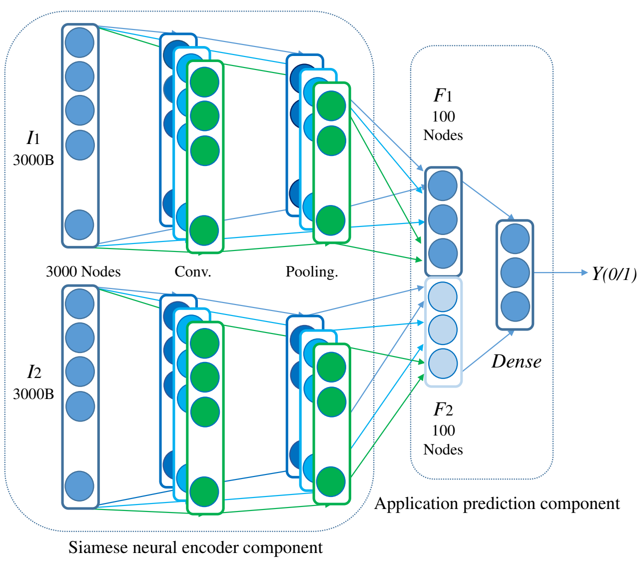

Figure 1: The proposed Siamese Convolutional Network model. The model takes two feature vectors and outputs a binary result.

2 DESIGN

Our model consists of two parts, as shown in Fig. 1, a Siamese neural encoder component and an Application prediction component. The model takes two flows of encrypted HTTP2 traffic as an input and determines whether or not they belong to the same application. Our objective is to use the Siamese neural encoder component to obtain a low dimensional representation of the two input traffic flows. Then the Application prediction component will suggest whether or not the two flows belong to the same application.

[1]https://drive.google.com/open?id=1CHKcWotJg_jjE2HH6g-Z1lfr1RBBy0LO

2.1 Siamese Neural Encoder Component

This component takes the feature vectors of two encrypted HTTP2 traffic flows as input. This encoder component contains two Siamese networks that share the same weights and biases. Each network starts with a layer of 3000 neuron nodes to take the input feature vectors I_1 and I_2. They are connected with fully connected one-dimensional convolutional layers respectively. The first convolutional layer has 64 filters with a kernel size of 3. The second convolutional layer has 32 filters with a kernel size of 3. Both layers have ReLU [4] as non-linearity. A Max-Pooling layer is added after convolutional layers to combine and reduce the spatial size of the low dimensional representation. The Siamese neural encoder component flattens the result of the Max-Pooling layer and outputs two vectors, F_1 and F_2, representing a low dimensional feature of the two input flows.

2.2 Application Prediction Component

Vector F_1 and F_2 from the previous component are concatenated to form a one-dimensional vector in this component. The concatenated vector has a size of 200 Bytes. The two fully connected dense layers are cascaded. The first layer has 100 nodes for each result vector from the Siamese neural encoder component. The second layer has 1 node. The first layer has ReLU as non-linearity. The second layer has Sigmoid as non-linearity.

3 EVALUATION

3.1 Implementation Details

To evaluate the approach we proposed, we trained and test our model on the dataset we collected. We only use 40 flows for each application to show the effectiveness of our model even with a small dataset. For each training batch, we randomly selected 200 input vector pairs for training. Half of the input vector pairs have their two traffic flows belonging to the same applications. The other 100 input vector pairs have their two traffic flows belonging to different applications. We also generate 20 pairs for the test set in each epoch. We run the training processing for 50 epochs. Finally, we generate 200 pairs of flows as a validation set to evaluate the model after training. The input vector is accumulated by concatenating the encrypted packet payload of a flow. If a flow does not contain sufficient byte streams, we append zeros to the end of the vector. We add 6 consecutive numbers of "FF" between any two payloads to serve as a boundary between them. This pattern of 6 "FF" is chosen because it never appears in any packet payloads. With a trade-off of accuracy and efficiency, an input vector size of 3000-bytes is chosen.

3.2 Preliminary Results

We performed an experiment based on the architecture and parameter we discussed above. The test was run on the Intel(R) Xeon(R) CPU E5-2690 v4 @ 2.60GHz CPU, 16GB Memory ,and a GeForce GTX 1080 Graphics Processing Unit. On the dataset we collected, our approach obtained a 92% accuracy using only 40 flows for each application. The time spent on testing each flow is 5ms on average. This computation time is much smaller than the duration of a whole traffic flow, which is typically longer than 1s.

REFERENCES

[1] Alexa 2020. *Alexa - Top sites*. Retrieved March 31, 2020 from https://www.alexa.com/topsites

[2] HTTP/2 2020. *Usage statistics of HTTP/2 for websites*. Retrieved April 3, 2020 from https://w3techs.com/technologies/details/ce-http2

[3] HTTPS 2020. *HTTPS encryption on the web*. Retrieved March 30, 2020 from https://transparencyreport.google.com/https/overview

[4] Vinod Nair and Geoffrey E. Hinton. 2010. Rectified Linear Units Improve Restricted Boltzmann Machines. In *ICML*. 807–814. https://icml.cc/Conferences/2010/papers/432.pdf

[5] S. Rezaei and X. Liu. 2019. Deep Learning for Encrypted Traffic Classification: An Overview. *IEEE Communications Magazine* 57, 5 (2019), 76–81.

[6] T. Shapira and Y. Shavitt. 2019. FlowPic: Encrypted Internet Traffic Classification is as Easy as Image Recognition. In *IEEE INFOCOM 2019 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. 680–687.